Aman Awasthi

Indian Institute of Technology, Bombay
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Personal Information

Nationality : **Indian**.

Sex : **Male**.

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Education

2018 -present **Ph.D. under the supervision of Prof. Subhabrata Dhar**, *Indian Institute of Technology, Bombay*, Mumbai, India.

2014–2016 **M.Sc. (Physics)**, *Chhatrapati Shahu ji Maharaj University*, Kanpur, Uttar Pradesh.

2011–2014 **B.Sc. (Physics, mathemarics, Chemistry)**, *Chhatrapati Shahu ji Maharaj University*, Kanpur, Uttar Pradesh.

Projects

2019 **Course work Project**, *Indian Institute of Technology, Bombay*, Mumbai, India.

Title: **Constraints on novel particles from stellar physics**.

2019 **Credit Seminar Project**, *Indian Institute of Technology, Bombay*, Mumbai, India.

Title: **Stellar Evolution**.

Research Interest

- Stellar Physics
- Astrophysics
- Helioseismology
- Cosmology
- Gravitational Waves
- Microlensing

Publication List

- *Detection prospects of solar g-modes with LISA.*
  Aman Awasthi ,
  Journal of Subatomic Particles and Cosmology (Under review) [arXiv:2602.18385]

Ongoing Research Projects

- **Revisiting the Hubble Constant from Supernova Data**
  Ongoing analysis of supernova-based measurements of the Hubble constant, with the aim of reassessing systematic uncertainties contributing to the current tension between early- and late-Universe determinations. This work is being conducted in collaboration, and the manuscript is currently in preparation.

- **Gravitational Microlensing**
  Developing a data processing pipeline for Roman Space Telescope observations to study gravitational microlensing events and investigate the underlying source characteristics.

- **Other Astrophysics ongoing projects**
  Other efforts are to study the population synthesis problem with the *binary_c* code and the application of LISA sensitivity on astrophysical sources.

Achievements & Rewards

2017 **Senior Research Fellowship**, *Joint CSIR-UGC NET (National Eligibility Test jointly held by Centre for Scientific & Industrial Research and University Grants Commission).*

2017 **GATE**, *Graduate Aptitute Test in Engineering* .

Professional Experience

Seminars Given

2018 **Stellar Evolution**, Indian Institute of Technology, Bombay,India.

Schools Attended

2018-Dec **GROWTH Winter School**, Indian Institute of Technology, Bombay,India, [Website].

2018-Dec **IITB-ICTP Workshop on Neutrino Physics**, Indian Institute of Technology, Bombay,India, [Website] .

2019-Jan **COSMOLOGY - THE NEXT DECADE**, International Centre for Theoretical Sciences, Banglore, India, [Website] .

2019-Oct **International Workshop on Frontier in High Energy Physics, FHEP 2019**, Indian Institute of Technology Hyderabad, University of Hyderabad, Hyderabad India , [Website] .

2020-Jan **QCD with Electron Ion Collider (QEIC)**, Indian Institute of Technology, Bombay,India, [Website].

2020-Nov **Less Travelled Path of Dark Matter: Axions and Primordial Black Holes (ONLINE)**, International Centre for Theoretical Sciences, Banglore, India, [Website] .

2021-Jan **School on Black Holes and Gravitational Waves (Online)**, Centre for String, Gravitation and Cosmology (CoE), Department of Physics, IIT Madras, India .

2021-May **Polarized Radiation near Supermassive Black Holes (Online)**, Princeton Center for Theoretical Science(PCTS), Princeton University,New Jersey, United States.

2021-May **Phenomenology 2021 Symposium (Online)**, University of Pittsburgh,Pittsburgh, Pennsylvania, United States.

2022-May **First IAGRG School on Gravitation and Cosmology (Online)**, Chennai Mathematical Institute, Chennai, India, [Website].

2023-March **Less Travelled Path to the Dark Universe**, International Centre for Theoretical Sciences, Banglore, India, [Website].

2024-Sept **Pune-Mumbai Cosmology and Astro- Particle Meeting**, Tata Institute of Fundamental Research, Mumbai, India, [Website].

2025-Feb **Pune-Mumbai Cosmology and Astro- Particle Meeting**, Indian Institute of Technology Bombay, Mumbai, India, [Website].

Teaching Experience

2020 Spring **Teaching assistant**, *For the course "Group Theory Methods in Physicists" at National Programme on Technology Enhanced Learning (NPTEL)*, The course is also publically available on , [Youtube link].

2019-2020 **Teaching assistant**, *For the course "BTech Lab I" at Indian Institute of Technology, Bombay, Dept. of Physics.*

Volunteer Experience

Science Outreach Volunteer

Technovanza 2019, Veermata Jijabai Technological Institute, Mumbai, India, *Dec – 2019*.

Explained scientific principles related to gravitational waves from astrophysical objects to students and their parents at local science exhibitions. Encouraged curiosity and promoted scientific literacy.

Volunteer on LIGO booth

Techfest 2020, Indian Institute of Technology Bombay, Mumbai, India, *Dec – 2019*.

Explained scientific principles related to gravitational waves from astrophysical objects to students. Interacted with students and explained them the science behind observations

Volunteer in International conference

2026 IITB-CFNS-CTEQ School on Perturbative QCD for EIC, Indian Institute of Technology Bombay, Mumbai, India, *Feb – 2026*.

Assisted in conference logistics, including registration, participant coordination, event scheduling, and food management. Contributed to smooth day-to-day operations.

Computer Skills

Operating systems: **Linux, MacOS**.

Languages: **FORTRAN, pyhton,**.

Packages: **Mathematica, MESA, GYRE, ADIPLS** .

Text Editors: **TeXStudio, Overleaf**.

LaTeX: **Proficient in LaTeX for typesetting, including packages such as `amsmath`, `tikz`, `bibtex`, and `graphicx`**.

Microsoft Office: **Word, Excel, PowerPoint (Advanced)**.

Languages Known

English, Hindi.